CAD Portfolio Highlights

Full Portfolio: https://github.com/jchen1231/CAD_Portfolio

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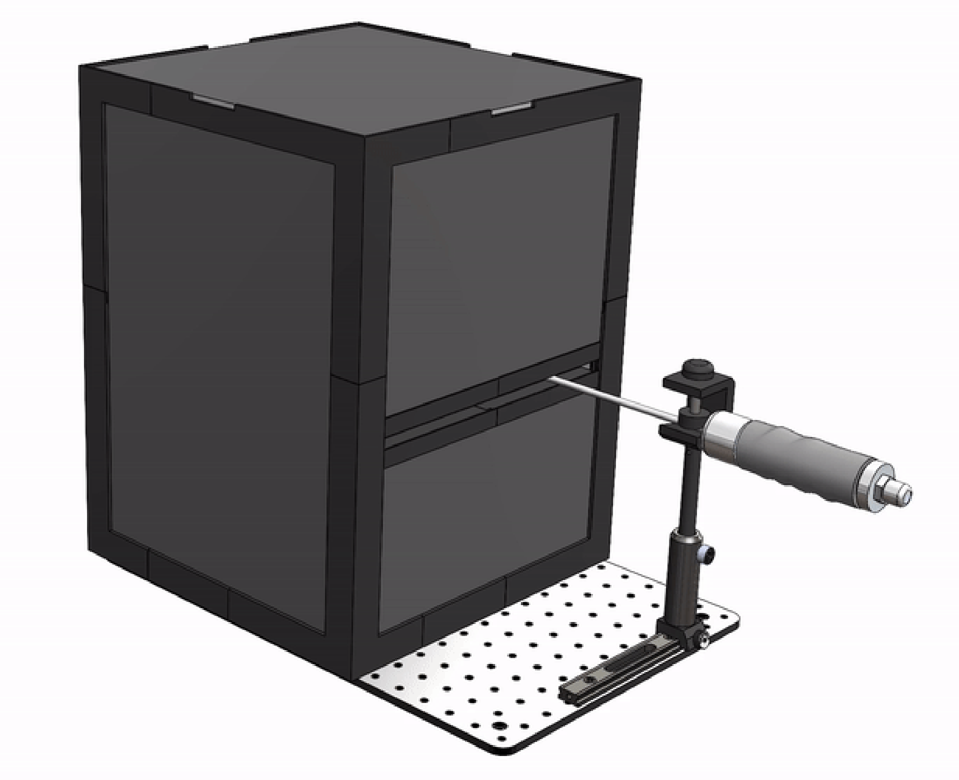


Project 1: Portable Testing Chamber

Mueller Lab

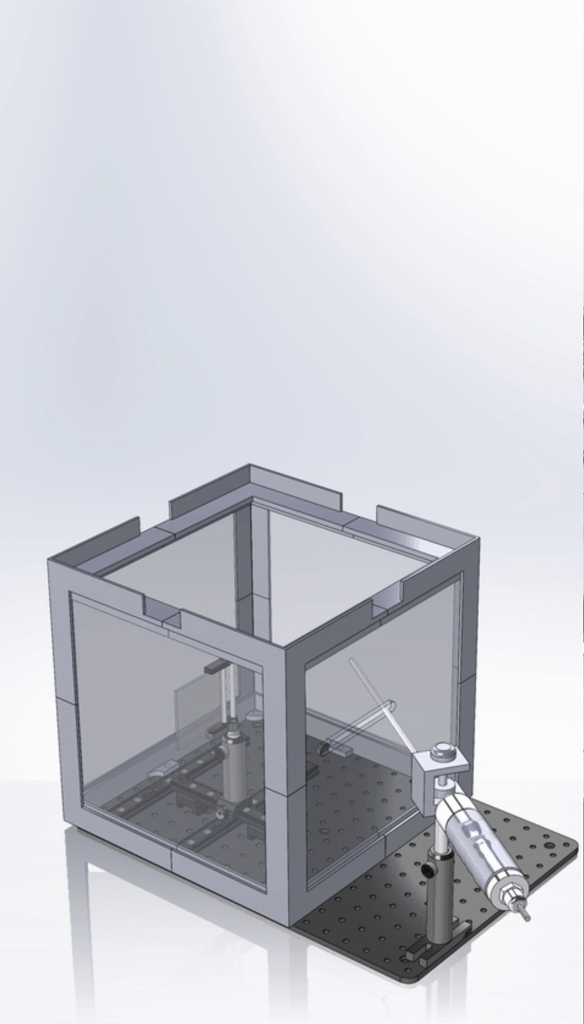
CAD models designed by Jason Chen

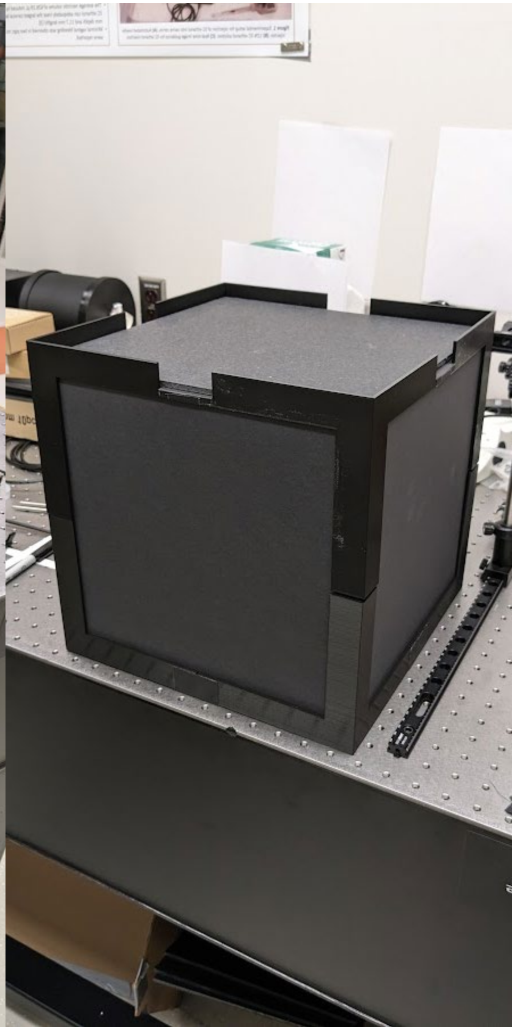
Fall 2021 – Fall 2023

Portable Testing Chamber v1 Model

Provide accessible laparoscope and camera characterization modalities, especially in low-resource or field-deployable settings.

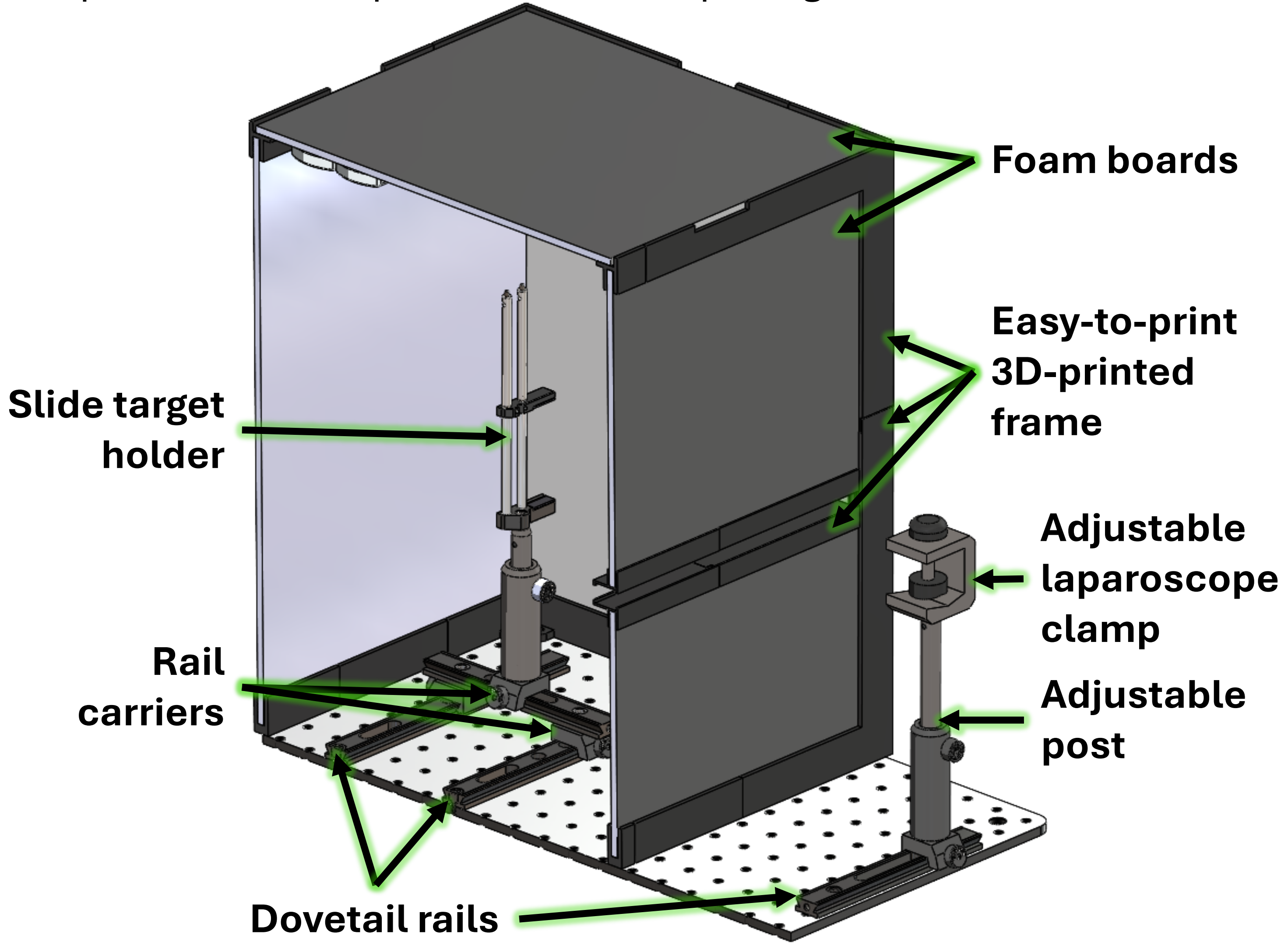
- Goals:
    - Compatible with a variety of laparoscopes and imaging test targets
    - Accessible components that can be easily sourced worldwide
    - Low-cost compared to commercial options
    - Easy to assemble, disassemble, and replace
    - Designed to fit within a standard 220 × 220 × 250 mm 3D printer build volume
- Tools used:
    - SolidWorks
    - Ender 3 Pro 3D-printer
    - Allen keys
    - Band saw
    - Luxmeter

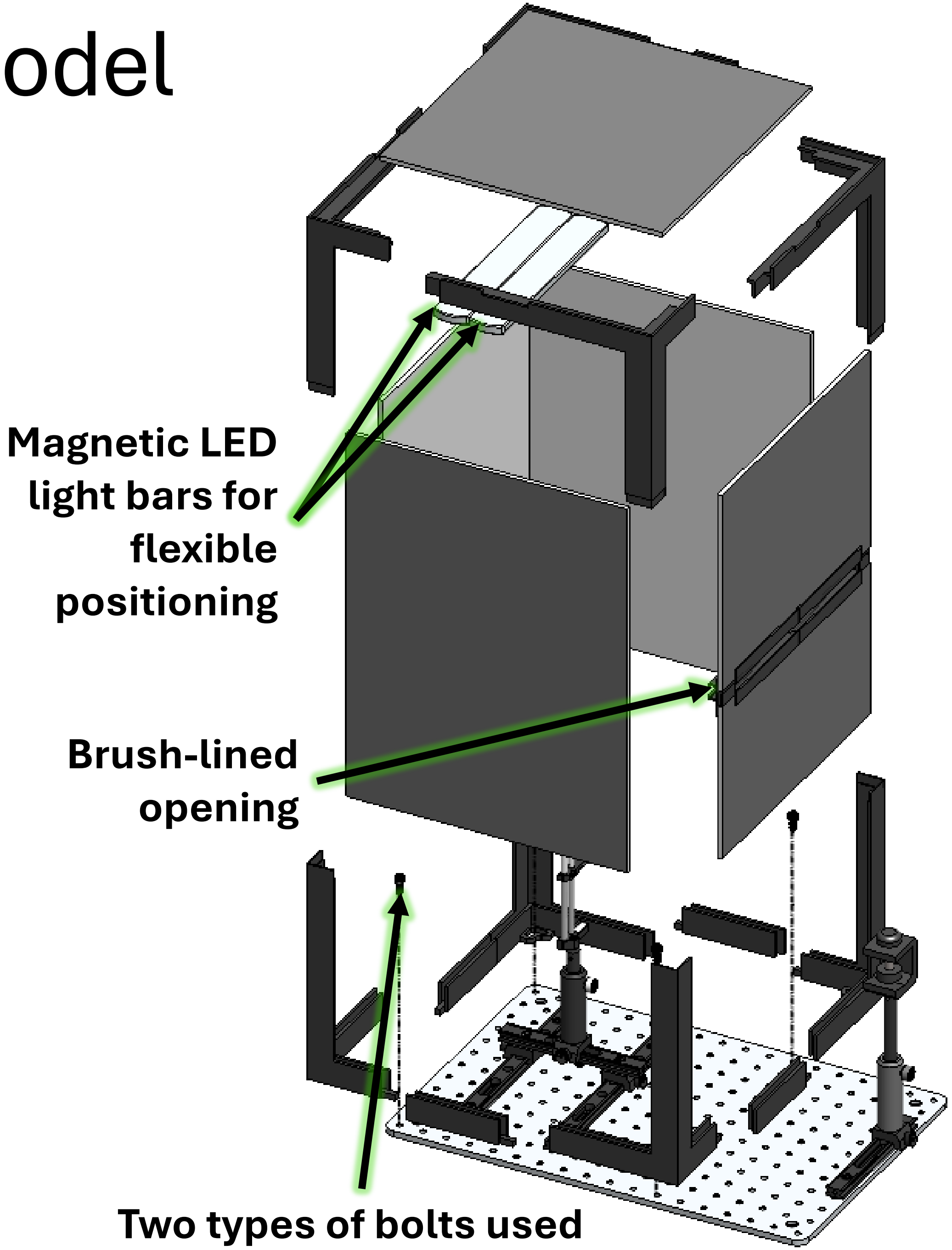




Portable Testing Chamber v4 Model

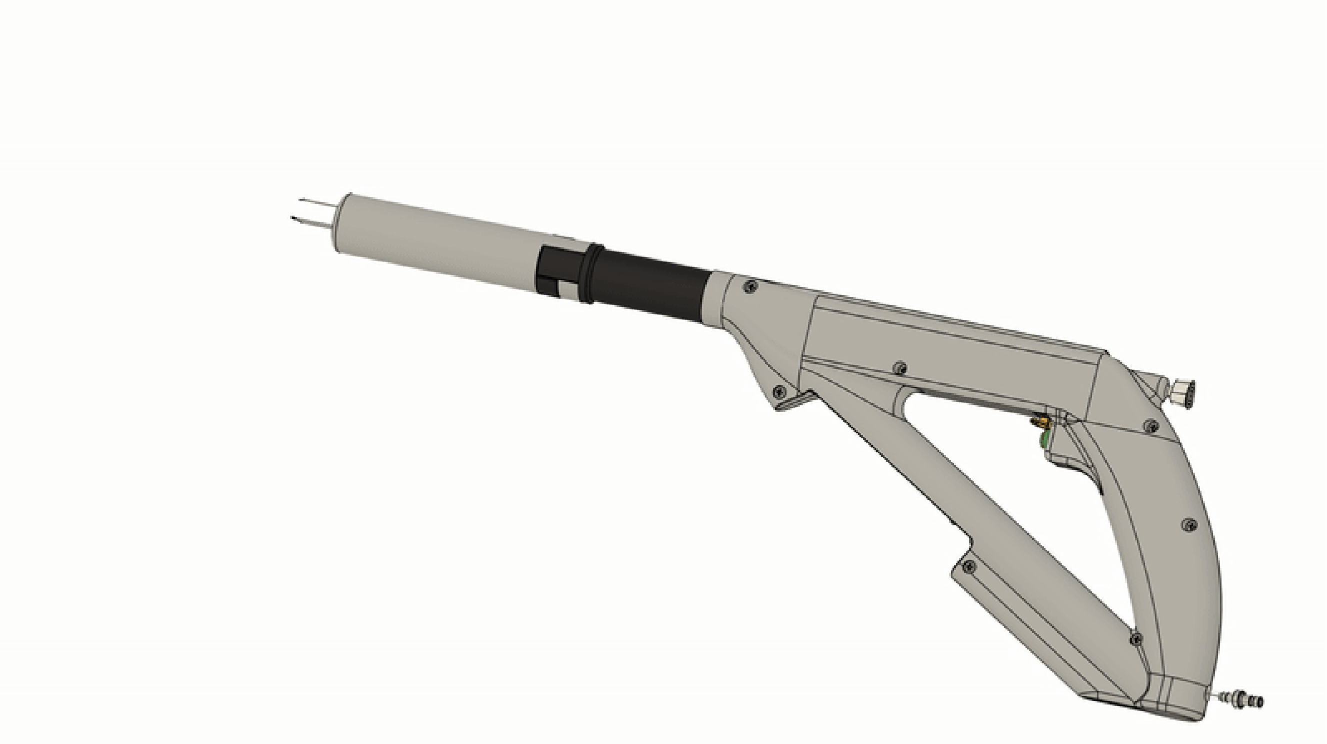
Updated to larger size to obtain more even lighting, added more versatile components, and improved chamber opening, and lowered costs.





Portable Testing Chamber Prototype

1. We ensured that our desired camera working distances (30–100 mm) were accommodated, and the PTC remained compatible with our image analysis targets, enabling consistent evaluation and effective optimization after each design iteration.
2. The brush-lined opening makes the PTC compatible with various laparoscopes, including our lab’s 5 mm, 10 mm, and 30-degree models.
3. The enclosure mainly consists of PLA, foam, and paper, which costs less than 25% of alternatives such as Thorlabs’ XE25C7/M.
4. We strategically split the frame into multiple components that could be printed on any hobbyist 3D-printer such as the Ender 3 Pro.
5. We designed disposable compact plastic nuts that could be tapped by bolts, as our desired nut dimensions were not commercially available.



Project 2: Auto-injector

Robert E. Fischell Institute for Biomedical Devices and Mueller Lab

CAD models by David Garvey and Jason Chen

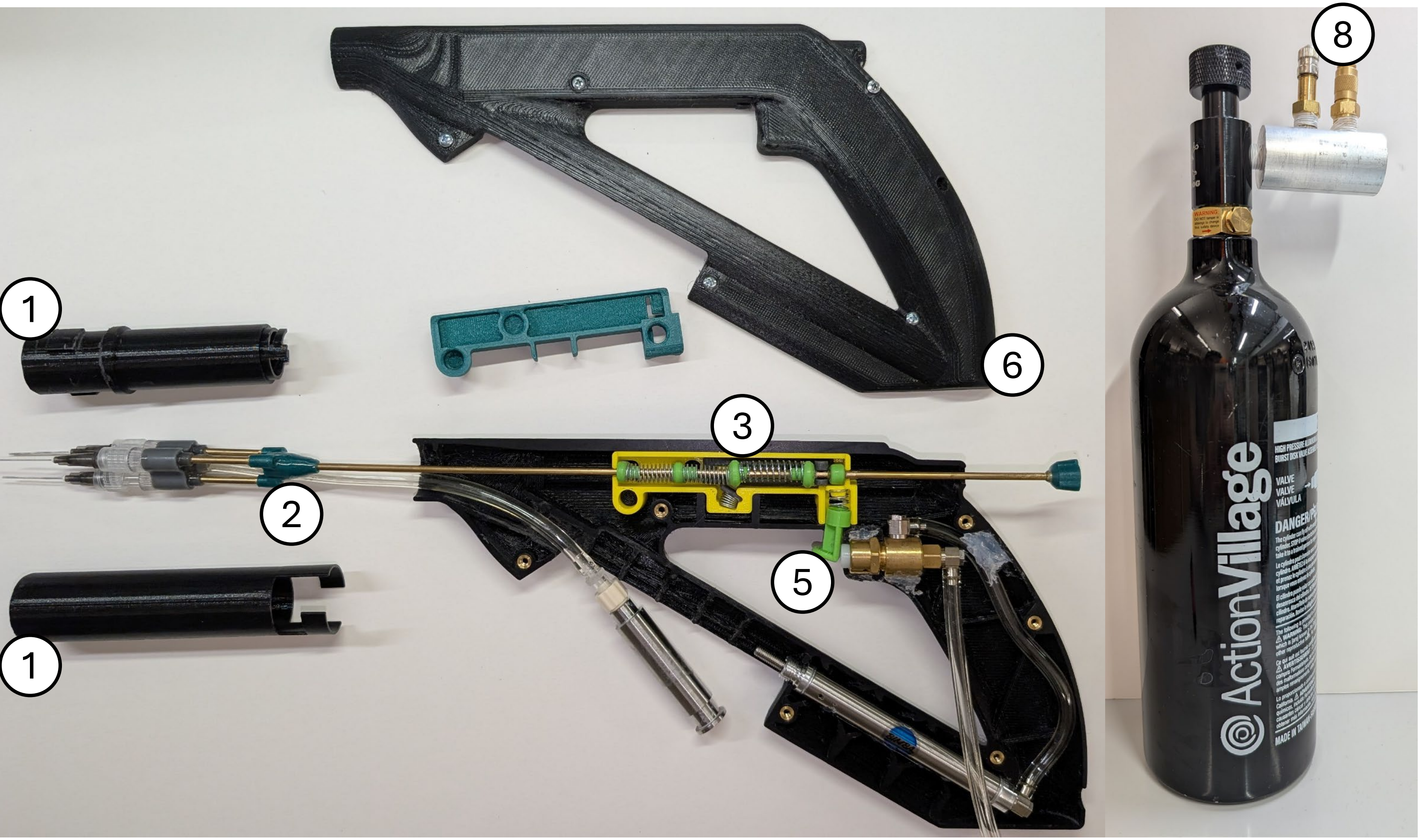
Fall 2023 – Spring 2024

Auto-injector – Legacy Prototype

Develop an easy-to-use, low-cost, triple-needle injector device that delivers three simultaneous deposits of a tumor-ablating therapeutic locally to create a larger area of tissue necrosis in one injection.

- Goals:
  - Easy to use and repair
  - Retracts needles to prevent injury
  - Injects viscous ECE solution in a timely manner
  - Compatible with a speculum
  - Reusable and low-cost
- Tools used:
  - Fusion 360
  - Prusa MK3S 3D-printer
  - Formlabs Form 3 SLA 3D-printer
  - Heat staking machine
  - Tap and die set
  - Lathe
  - Band saw
  - Belt sander
  - Drill press

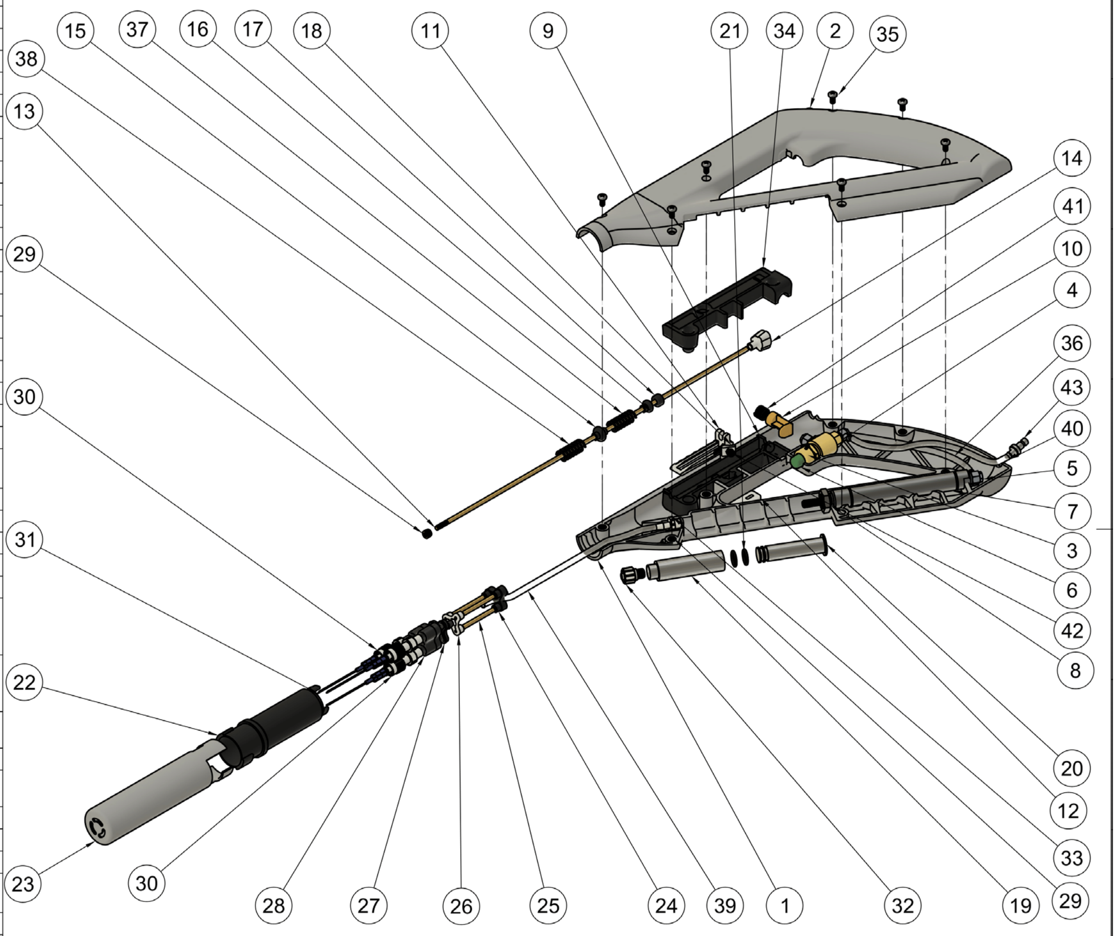
Auto-injector – Updated Prototype



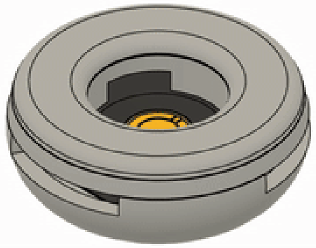
1. Designed detachable and speculum-compatible needle covers.
2. The distal triple-needle assembly is detachable through an added threaded insert; the long rod and springs can now remain in place.
3. The retraction mechanism now functions reliably after minimizing friction, refining latch designs, and integrating stronger springs.
4. The retraction housing is fully modular and is held in place using alignment holes/ribs, friction, and pressure fits.
5. Adding a trigger button latch removes the need to constantly hold down the button.
6. Maintained a housing design compatible with injection molding.
7. Reduced the number of screws and other pneumatic components.
8. Air canister compatible with both bike pump and air compressor.

Auto-injector – Updated Prototype

Parts List			
Item	Qty	Part Number	Material
1	1	Right_half	Steel
2	1	Left_half	Steel
3	1	Button_pc-2w	Steel
4	1	Valve_mav-2_1	Steel
5	3	Ut0-4-pkg_1	Steel
6	1	Trigger_button_config_1	ABS Plastic
7	1	6498K116_Self_Retracting_Air_Cylinder	Steel
8	1	6498K116_Air_Cylinder_nut	Steel
9	1	Retraction_housing_bot_rev_H	----
10	1	Trigger_brake_config_2	ABS Plastic
11	1	Retraction_slider_config_1	ABS Plastic
12	1	Retraction_trigger_rev_E	ABS Plastic
13	1	Dia_3mm_rod_config_1	Brass
14	1	Trigger_rod_button_rev_E	ABS Plastic
15	1	Spring_spacers_config_1	ABS Plastic
16	1	Spring_spacers_config_2	ABS Plastic
17	1	Retraction_slider_to_rod_rev_D_v4	ABS Plastic
18	1	Retraction_door_rev_H	ABS Plastic
19	1	Reusable_Syringe_Body	Aluminum 6061
20	1	Reusable_Syringe_Plunger	Aluminum 6061
21	2	9263K622_Chemical-Resistant Viton Fluoroelastomer O-Ring	Rubber, Nitrile
22	1	Proximal_needle_cover_rev_D	ABS Plastic
23	1	Distal_needle_cover_rev_D	----
24	1	Needles_3_to_1_adapter	ABS Plastic
25	3	Dia_3mm_rod_config_2	Brass
26	1	Needles_3_to_tip_adapter	ABS Plastic
27	1	Needles_holder_back	Tough 2000 (with Formlabs SLA 3D Printers)
28	1	Needles_holder_front	Tough 2000 (with Formlabs SLA 3D Printers)
29	8	94180A331_Tapered Heat-Set Inserts for Plastic	Brass
30	3	90245	ABS Plastic
31	3	5791N35_Sterile Stainless Steel Dispensing Needle with Luer Lock Connection	----
32	1	51525K221_Plastic Quick-Turn Tube Coupling	Nylon 6
33	1	51525K213_Plastic Quick-Turn Tube Coupling	Nylon 6
34	1	Retraction_housing_top_rev_D	ABS Plastic
35	7	90116A151_316 Stainless Steel Pan Head Phillips Screws	Steel
36	1	Sample_tubing_1	PVC, Flexible
37	1	Retraction_spring_2	Steel
38	1	Retraction_spring_3	Steel
39	1	Sample_tubing_3	PVC, Flexible
40	1	Sample_tubing_2	PVC, Flexible
41	1	Retraction_spring_4	Steel
42	1	Retraction_spring_1	Steel
43	1	QDC-102-E-1332	Stainless Steel AISI 304



Dept.	Technical reference	Created by Jason Chen 5/31/2025	Approved by
SCALE 1:2 UNITS: mm		Document type	Document status
		Title auto_injector_rev_D	DWG No.
			Rev. / Date of issue / Sheet 1/1



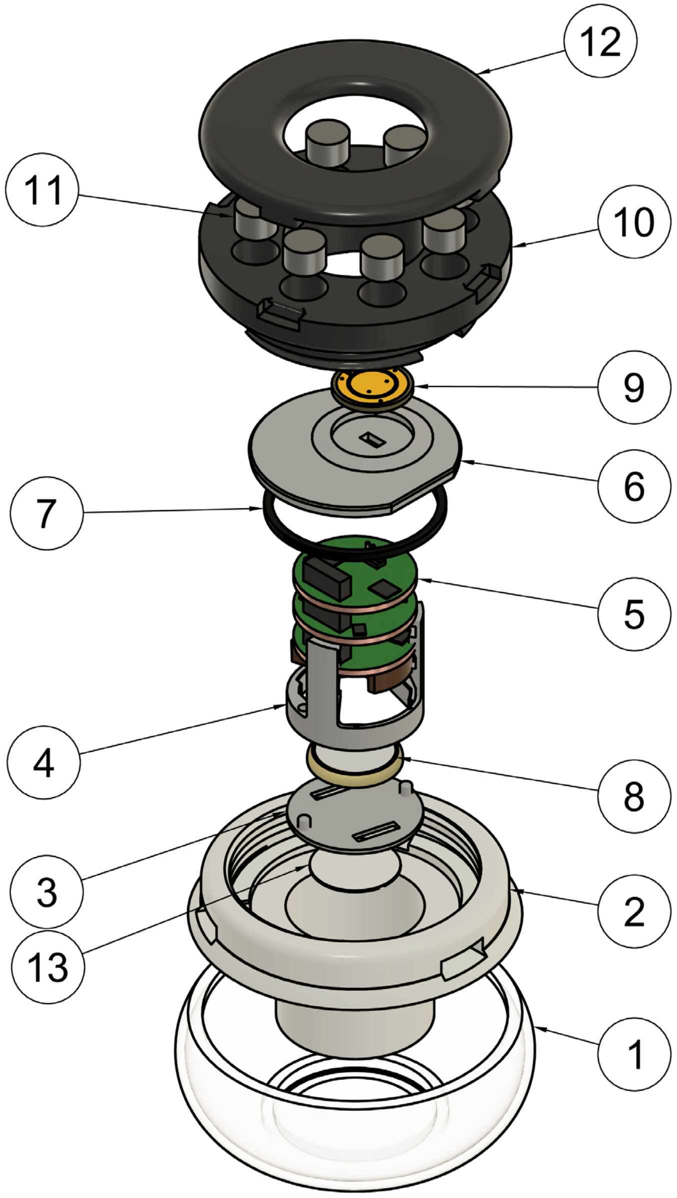
Project 3: Bobblestat Biosensor

Robert E. Fischell Institute for Biomedical Devices

CAD models by Quinn Burke and Jason Chen

Spring 2024 – Current

Bobblestat revision 1 Prototype

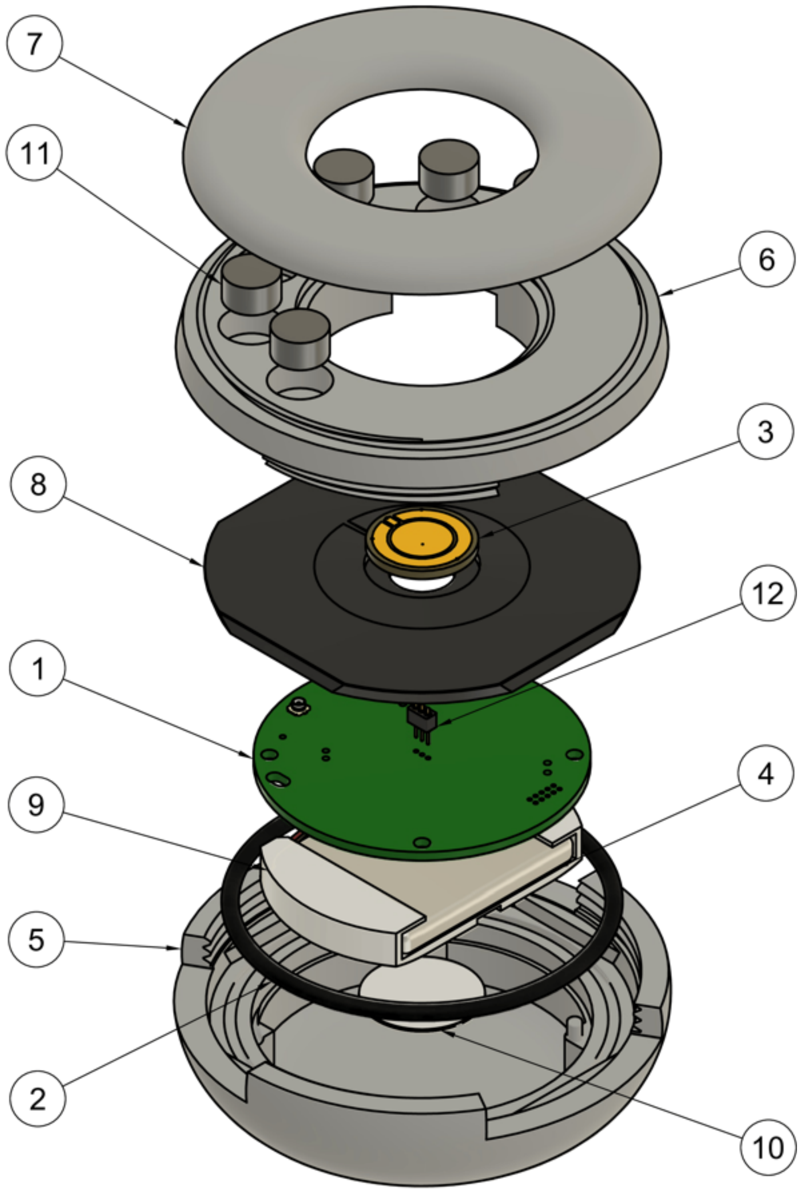


PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	HOUSING_FLEXIBLE_EXTERIOR	POLYETHYLENE, LOW DENSITY
2	1	HOUSING_HARD_INTERIOR	TOUGH 2000 (WITH FORMLABS SLA 3D PRINTERS)
3	1	PCB_SCAFFOLD_BOT	ABS PLASTIC
4	1	PCB_SCAFFOLD_TOP	ABS PLASTIC
5	1	BOBBLESTAT_PCB_MODEL	----
6	1	ELECTRODE_SUBSTRATE	TOUGH 2000 (WITH FORMLABS SLA 3D PRINTERS)
7	1	4061T133_SQUARE-PROFILE OIL-RESISTANT BUNA-N O-RING	RUBBER, NITRILE
8	1	CR2032	----
9	1	ASSEMBLED 17MM_W3C4 REV H	----
10	1	LID_BOTTOM	TOUGH 2000 (WITH FORMLABS SLA 3D PRINTERS)
11	8	WEIGHTS_1G	STEEL
12	1	LID_TOP	TOUGH 2000 (WITH FORMLABS SLA 3D PRINTERS)
13	1	760308101214	----

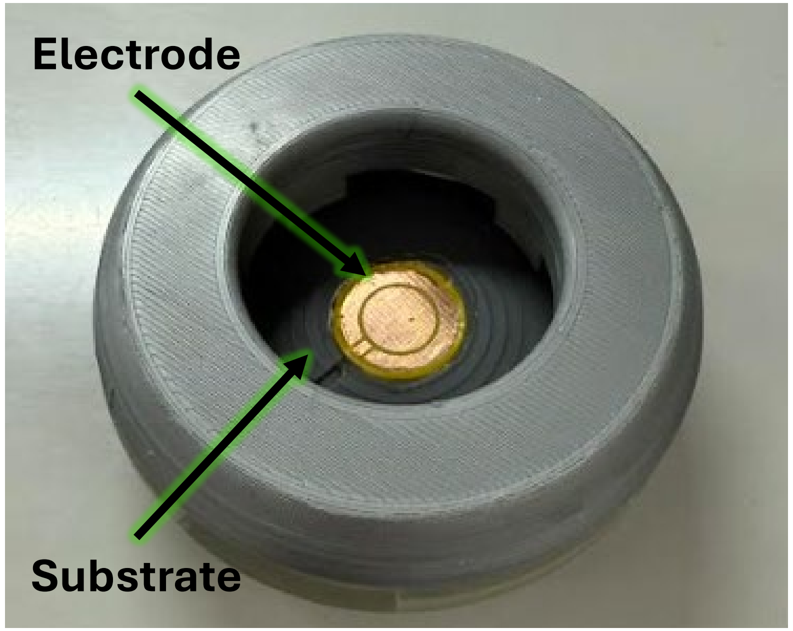
- Develop a watertight, modular, and buoyant biosensor that integrates potentiostat PCB and swappable electrodes.
- Goals:
  - Modular
  - Water-resistant
  - Adjustable buoyancy
  - Minimize trapping internal air bubbles
  - Optimize antenna orientation
- Tools used:
  - Fusion 360 and Fusion 360 Electronics Workspace
  - Formlabs Form 3 SLA 3D-printer
  - Prusa MK3S 3D-printer
  - Soldering station
  - Electroplating supplies (e.g., DC power supply)
  - Wet lab supplies (e.g., pipettes)
  - Brightfield and scanning electron microscope
  - Multimeter

*NOTE: Clear resin was used for this prototype.

Bobblestat revision 2 Prototype



Parts List			
Item	Qty	Part Number	Material
1	1	PCB	---
2	1	2418T158	Rubber, Nitrile
3	1	17mm_W3C4PP rev I	---
4	1	LiPo_Battery_Dummy	---
5	1	Flex_housing	Tough 2000 (with Formlabs SLA 3D Printers)
6	1	Cap_A	Tough 2000 (with Formlabs SLA 3D Printers)
7	1	Cap_B	Tough 2000 (with Formlabs SLA 3D Printers)
8	1	Substrate	Tough 2000 (with Formlabs SLA 3D Printers)
9	1	Battery_scaffold	ABS Plastic
10	1	760308101214	---
11	6	Weight_1g	Steel
12	1	Mill_max-854-22-003-10-004101	---



- Updated housing to accommodate a new condensed PCB form factor.
- Switched to POGO pins, which offer greater longevity and reduce assembly complexity but require more careful alignment.
- Reduced the number of weight slots to only what is needed for orientation.
- Converted snap-fit lid to screw-on lid to improve fit and durability.
- O-ring groove location moved from the electrode substrate (Item 5) to the housing side (Item 8), and it now features a dovetail cross-section to improve retention.
- Modified the substrate to be squarer, providing a more stable fit than the previous circular outline.
- Enlarged and realigned the degassing slots to facilitate air bubble escape.
- Added an alignment groove to the substrate to aid in electrode orientation.
- Ensures that the antenna always faces upward when floating for best signal quality.